A square-tessellation quotient below the claimed spectral bound

Disproof of Conjecture 00000007717

gaochengzhecpu

Abstract

The square tessellation, quotiented by the period subgroup $64\mathbb{Z} \times 4\mathbb{Z}$, has vertex degree four and four edges per cell. Its actual combinatorial Laplacian has a positive eigenvalue $2 - 2\cos(\pi/32) < 3/256$. Consequently its first nonzero eigenvalue is below $(3/16)/4^2$, contradicting the asserted constant $c = 3/16$. Lean verifies the quotient, connectivity, degree, cells, real eigenvector and least positive eigenvalue.

The admissible tessellation and its quotient

Take the ordinary planar square tessellation with vertex set $\mathbb{Z}^2$ and edges joining points whose difference is $(\pm 1, 0)$ or $(0, \pm 1)$. The subgroup $\Lambda = 64\mathbb{Z} \times 4\mathbb{Z}$ consists of periods of this tessellation. Its quotient graph has vertex set

$$V = (\mathbb{Z}/64\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z}).$$

Put $a = (1, 0)$ and $b = (0, 1)$. A vertex v is adjacent exactly to $v + a, v - a, v + b, v - b$. These are four distinct vertices, none equal to v, so the graph is simple and has degree exactly four. Horizontal and vertical steps connect every vertex to the origin. Every square cell projects to the four distinct vertices

$$v, \quad v + a, \quad v + a + b, \quad v + b,$$

with consecutive vertices adjacent, including the last to the first. Thus the source's bounds hold with $D = 4$ and $E = 4$.

The quotient map $\mathbb{Z}^2 \to V$ reduces the two coordinates modulo 64 and 4. It is surjective and its kernel is exactly Λ. Two quotient vertices are adjacent if and only if they have adjacent lifts. These facts are formalized, so the construction is an actual quotient of a tessellation 1-skeleton. The statement does not require using the full group of all translation symmetries or a smallest period cell. Our chosen subgroup is a period subgroup, and no such extra restriction is imposed here.

An exact small eigenvalue

For the combinatorial Laplacian, with no normalization,

$$(Lf)(i,j) = 4f(i,j) - f(i+1,j) - f(i-1,j) - f(i,j+1) - f(i,j-1).$$

Let $\theta = \pi/32$ and $f(i,j) = \cos(i\theta)$. This is well-defined modulo 64, independent of j, and nonzero since $f(0,0) = 1$. The cosine addition formula gives

$$Lf = (2 - 2\cos\theta)f.$$

Write $\lambda = 2 - 2\cos\theta$. Since $0 < \theta < \pi$, $\cos\theta < 1$ and $\lambda > 0$. The elementary inequality $\cos x \geq 1 - x^2/2$ gives

$$\lambda \leq \theta^2 = \frac{\pi^2}{1024} < \frac{12}{1024} = \frac{3}{256},$$

where $\pi^2 < 12$ is certified exactly in Lean from a rational upper bound for π.

The real Laplacian is positive semidefinite, so its nonzero eigenvalues are positive. In this finite graph, the set of positive eigenvalues is finite and nonempty, as it contains λ. It has a least element λ_1, and

$$0 < \lambda_1 \leq \lambda < \frac{3}{256} = \frac{3/16}{D^2} \qquad (D = 4).$$

Theorem 1. *The constant $c = 3/16$ asserted in Conjecture 00000007717 does not satisfy its claimed bound on all periodic tessellation cell graphs.*

Proof. The quotient above is connected, has $D = E = 4$, and has first nonzero eigenvalue strictly smaller than $(3/16)/4^2$. It therefore violates that bound. □

This finite example refutes the explicit value claimed for the optimal universal constant, hence the stated conjunction. It does not rely on a disconnected graph, weighted edges, growing degrees, or a normalized Laplacian. No claim to formalize an entire family of arbitrarily large quotients is needed.

Formalization and reproducibility

The graph is a Mathlib `SimpleGraph` on the actual finite type `ZMod 64` times `ZMod 4`. Theorems `project_kernel` and `quotient_adjacency` identify the quotient and its adjacency relation. Degree, connectivity and four distinct cell vertices are checked separately.

The proof first uses the actual additive character on $\mathbb{Z}/64\mathbb{Z}$ as a complex eigenvector. It identifies the eigenvalue with the real number λ and takes real parts to obtain a nonzero eigenvector of the actual real Mathlib Laplacian matrix. The theorem `spectral_gap_exists_and_small` uses the matrix linear map's actual eigenvalues and constructs their least positive element. The theorem `counterexample` bundles the graph properties and this strict bound. Positive semidefiniteness is also verified. A small scalar estimate is therefore never substituted for the spectral gap itself.

An independent exact-arithmetic check in `verify.py` builds all 256 vertices and their undirected edges. For the mean-zero triangular-wave function $h(i,j) = \min(i, 64-i) - 16$, it obtains

$$\sum_v h(v) = 0, \qquad \sum_v h(v)^2 = 21888, \qquad \sum_{\{u,v\}\in E} (h(u)-h(v))^2 = 256.$$

Its Rayleigh quotient is $2/171 < 3/256$. This is supplementary verification; Lean uses the explicit trigonometric eigenvector instead.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Run `python verify.py` and `tectonic main.tex` for the auxiliary check and document. Fresh full build, strict Lean, axiom audit, source provenance and PDF verification records are included. Only standard foundational axioms occur; no proof placeholders, custom axioms or external computation oracles are used.

An agent developed the proof and supplied a self-review. The parent separately reviewed its mathematical scope, complete Lean source and final artifacts. No external independent review or novelty is claimed for the elementary Fourier calculation.

References

[1] The Last Math Competition, Conjecture 00000007717. The exact bilingual statement and commit-pinned provenance are included as `SOURCE.md` and its verification record.